

PROJECT REPORT

Title

AI-Powered Ticket-to-FAQ Pipeline Using Gemini AI

Abstract

Customer support teams receive thousands of tickets, many of which describe recurring issues. Manually analyzing these tickets and creating Frequently Asked Questions (FAQs) is time-consuming and inefficient. This project automates the process by clustering similar closed support tickets and generating FAQ drafts using Google's Gemini AI. The generated FAQs are reviewed by administrators before being published to the knowledge base. The system improves support efficiency, reduces repetitive work, and helps organizations maintain an up-to-date FAQ repository.

Introduction

Organizations often face repetitive customer support issues. Support agents repeatedly answer similar questions, leading to increased workload and slower response times. The Ticket-to-FAQ Pipeline addresses this challenge by automatically identifying recurring issues from historical tickets and generating reusable FAQs.

Problem Statement

Manual creation of FAQs from support tickets is:

- Time-consuming
- Error-prone
- Difficult to scale
- Dependent on support team effort

An automated solution is required to convert recurring support tickets into structured FAQs.

Objectives

1. Analyze closed support tickets.
2. Identify recurring issues.
3. Cluster similar tickets together.
4. Generate FAQ drafts using AI.
5. Enable admin review and approval.
6. Publish approved FAQs to the knowledge base.

System Architecture

Customer Portal



Support Panel



Closed Tickets Database



Data Processing & Clustering



Gemini AI FAQ Generation



Admin Review



Published FAQ Repository

Technologies Used

Frontend

- Streamlit

Backend

- Python

Database

- SQLite

AI Model

- Google Gemini API

Libraries

- Pandas
- Scikit-learn
- Streamlit
- SQLite3

Version Control

- GitHub

Modules

1. Customer Portal

Allows users to submit support tickets.

2. Support Panel

Support staff can review and close tickets with resolutions.

3. Closed Tickets Module

Stores resolved tickets for analysis.

4. Clustering Module

Groups similar tickets into clusters based on issue similarity.

5. FAQ Generation Module

Uses Gemini AI to generate FAQ drafts from ticket clusters.

6. Admin Review Module

Allows administrators to approve, reject, or edit generated FAQs.

7. Published FAQ Module

Stores approved FAQs and supports export functionality.

Workflow

Step 1: User submits a support ticket.

Step 2: Support agent resolves and closes the ticket.

Step 3: Closed tickets are processed.

Step 4: Similar tickets are grouped into clusters.

Step 5: Gemini AI generates FAQ drafts.

Step 6: Admin reviews generated FAQs.

Step 7: Approved FAQs are published.

Features

- Ticket management
- Automated clustering
- AI-powered FAQ generation
- Admin approval workflow
- FAQ publishing
- Dashboard analytics
- Export functionality

Advantages

- Reduces manual effort
- Improves support efficiency
- Detects recurring problems
- Creates knowledge base automatically
- Easy-to-use interface

Results

The system successfully:

- Loaded closed support tickets
- Clustered recurring issues
- Generated FAQ drafts using Gemini AI
- Supported admin review and approval
- Published approved FAQs

Future Enhancements

- Real-time ticket integration
- Multi-language FAQ generation
- Vector database support
- Advanced semantic search
- Cloud deployment

Conclusion

The AI-Powered Ticket-to-FAQ Pipeline demonstrates how Artificial Intelligence and Machine Learning can automate knowledge management in customer support environments. The project successfully converts recurring support tickets into meaningful FAQs, reducing support workload and improving customer experience.

